

Heat Recovery Units (HRUs): Process, Flow & Optimization Overview

Purpose:

Heat Recovery Units (HRUs) recover waste heat from exhaust air or fluids and transfer it to incoming fresh air or process water. This improves energy efficiency and reduces heating/cooling loads, especially in HVAC systems, manufacturing, and data centers.

System Start-to-Finish Flow:

1. **Capture of Waste Heat**
 - HRUs intercept exhaust air (from AHUs, kitchens, labs) or warm process fluids.
 - Air-to-air or liquid-to-liquid exchangers extract sensible or latent heat.
2. **Heat Transfer Mechanism**
 - Common types: plate exchangers, rotary wheels, run-around coils, shell-and-tube.
 - No cross-contamination between exhaust and intake air.
3. **Delivery to Fresh Supply or Process Input**
 - Heated air or water preconditions incoming supply for AHUs or boilers.
 - Lowers need for direct heating sources.
4. **Monitoring & Controls**
 - Temp sensors and control valves regulate recovery rate.
 - Differential pressure sensors detect fouling or clogging in exchangers.

Key Components & Definitions:

- **Run-Around Coil**: Closed loop using glycol or water to transfer heat between coils.
- **Rotary Wheel**: Slowly rotating core transfers heat between opposing air streams.
- **Plate Heat Exchanger**: Corrugated metal plates transfer heat without mixing fluids.
- **Energy Recovery Ventilator (ERV)**: Combines heat and moisture recovery.
- **Efficiency Ratio**: Measures % of exhaust energy recovered (typically 50–80%).

Example Flow:

> A data center installs a plate-type HRU to recover heat from server exhaust and preheat incoming winter air. The system raises inlet temps from 32°F to 58°F, reducing gas heating usage by 30% and preventing coil freeze-ups.